Global warming and choices

Today I can… ☐ I can put cause before effect. ☐ I can say how gases add to warming. ☐ I can compare a cut in emissions with help for an impact.

Staff answers

GROW_SCI_A2_W04 · GROW Science teacher note

Staff-facing; review only. Verbatim SoW objective: Describe causes and effects of global warming and how to reduce our impact.

Source: GROW scheme of work, Autumn 2 week 4. Pupil goal: explain a cause and an effect of global warming, then compare two kinds of action.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40. One-minute resets are inside We do 2, Review and Exit. Success: I can put cause before effect., I can say how gases add to warming., I can compare a cut in emissions with help for an impact.

Retrieval and I-do check

When | Question | Answer | SoW cell

last lesson | Name one fact about the Solar System and its class card. | The Sun is a star, on card A; or Earth goes round the Sun, on card B. | GROW scheme of work, Autumn 2 week 3

2-4 weeks | Why does one place have night? | Earth turns so that place faces away from sunlight. | GROW scheme of work, Autumn 2 week 1

last term | Name one way friction can affect motion. | It can slow a moving object. | GROW scheme of work, Autumn 1 week 3

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Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: Point to the cause card: burning fuel or a hotter day? Answer: Burning fuel is the cause card; a hotter day may be an effect. All pupils show by point, card or board.

Words (same as page and organiser):

Word | Tier | Simple definition | Example sentence

climate | Tier 3 | the pattern of weather over many years | Climate is not the weather on one day.

carbon dioxide | Tier 3 | a gas added to the air when fossil fuels burn | Burning coal adds carbon dioxide to the air.

greenhouse gas | Tier 3 | a gas in the air that slows heat leaving Earth | Carbon dioxide is a greenhouse gas.

impact | Tier 2 | an effect on people or places | More heat can have an impact on people.

Faded example, tiered task and stretch

I do full: “Burn fossil fuel → add carbon dioxide → more heat stays → heat risks. Cut fuel use to cut emissions; plan shade to help with heat.” We do part-done: “Burn fuel → add carbon dioxide → → heat risks.” Removed step: Explain that greenhouse gases slow heat leaving Earth. Independent: no filled step. Next fade: Next time omit the gas card and ask pupils to choose both middle links from a mixed set.

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We do F: Lay the first and last cards; point to each missing space before choosing a card. M: Explain two chain links. H: Explain why a flood defence does not remove carbon dioxide already added by burning fuel. Independent F: Order one cause and effect. M: Explain two chain links. H: Compare an emissions cut with an adaptation. Go deeper: Why might a council cut fuel use and plan flood defences? Name a limit of each. Named harder stretch — Two kinds of action: Choose a council goal, justify the first action and explain the problem still left.

The model, source or practical route is: warming-chain cards, ice and card observation trays. Two on-screen interactions are card sort and Which action helps with an impact rather than cutting emissions?; native keyboard controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Practical: Predict first: Will ice on dark and light card melt alike? Kit: Two equal ice pieces, Two trays, Dark card, Light card, Staff clock. Steps: Put equal ice pieces on cards in trays; Place both trays in the same light; Observe both at one set time; Record what you see. Say why this is not a climate model. Results table: Time | dark card ice | light card ice, Start | observation | observation, Later | observation | observation. Check against the school risk assessment and [SYS] guidance. Dry spills; use no heat source. This surface observation is not proof of the greenhouse effect.

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Hinge and decision

Question: A bus depot burns more diesel. Which immediate link belongs in its emissions chain?

Option | Correct? | Diagnosis

More carbon dioxide is released. | yes | Uses the new case accurately.

The extra carbon dioxide appears only when the depot has less daylight. | no | Makes the emissions depend on daylight instead of fuel burnt.

The depot warms first; that warmth makes its engines release carbon dioxide. | no | Reverses the chain: warming follows added greenhouse gases, rather than causing this exhaust.

Only local air pollution changes; the amount of carbon dioxide stays the same. | no | Confuses a local air-quality effect with the absence of carbon-dioxide emissions.

Rule: About 3 in 4 or more right → move on. Otherwise re-model with the named I-do example, then ask the parallel hinge. Parallel: Third case: a building burns less gas for heat. What changes first in this simplified chain? Answer: less carbon dioxide from burning that gas, all else equal.

Misconceptions:

Misconception | Spot | Correct

Less daylight controls whether diesel exhaust releases carbon dioxide. | Pupil chooses B, making emissions depend on daylight. | Trace the immediate fuel-burning step regardless of daylight.

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Staff answers continued

Diesel exhaust only changes local air quality and adds no carbon dioxide. | Pupil chooses D, treating local pollution as the only diesel effect. | Separate local air-quality effects from the added greenhouse gas.

[AWARD] Entry Level Science style bridge (classroom planning): Order and explain two links in an environmental cause-and-effect chain. This is a planning bridge, not a claimed unit outcome or code.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ?. At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ?. R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations

Broad area | EEF five-a-day strategy | Specific support

Cognition & Learning | scaffolding | Use four large chain cards with the first and last fixed, then remove a cue.

Communication & Interaction | cognitive/metacognitive strategies | Let partners rehearse an M/H because sentence while pointing to the chain.

SEMH | flexible grouping | Give a quiet choice of two card roles and a pause signal before the hinge.

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Staff answers continued

Sensory/Physical | technology | Provide large high-contrast cards and a stable board; accept voice capture of the ordering.

EAL (language, not SEN) | explicit instruction | Teach gas, cause and impact with arrows. Plan in any language; use M/H frames and keep the comparison challenge.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Use the held five-step prompt ladder in the TA brief. No physical prompt. Source-specific stage cues remain below.

At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Optional invitation bank, answers:

- F: Point to the cause card. → Burning fossil fuels.; Point to a possible effect. → More heat or a greater flood risk, in some places.

- M: What is added to the air when fuel burns? → Carbon dioxide.; What does a greenhouse gas do? → It slows heat loss from Earth.

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Staff answers continued

- H: How do emissions cuts and flood defences differ? → One limits the cause; the other helps with an impact.; Why does a hot day alone not prove a climate trend? → Climate is a pattern over many years.

Regulation: Agree a quiet pause signal and supervised place. Check in before hinge and Independent; point to the last completed step on re-entry. No evidence while a pupil is distressed.

lesson record: GROW_SCI_A2_W04 via staff device or projector only. Gatsby 4: Building energy adviser: help reduce fuel use and explain how that cuts emissions.

Historical source screening only; superseded by current actual-artifact reading reproduction in CHECKS. Reading gate: GROW 8–11; rough visible-body ARI grade 4.45; age guide 10.45 years; rough FK grade 4.27; 830 pupil-visible words excluding marked adult guidance and navigation. Screening estimate only. This is a rough screen, not an individual assessment.

Pupil-facing risk: Melted ice can leave a slippery spill.

Pupil-facing safety: Keep ice in trays. Wipe spills and use no heater.

SLT planning and evaluation

Lesson code: SCI-GROW-A2-W04

SoW reference: GROW scheme of work, Autumn 2 week 4

Exact SoW outcome: Describe causes and effects of global warming and how to reduce our impact.

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Staff answers continued

Evidence (scheme of work): [AWARD] [QUAL]; [AWARD] [QUAL]. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: [AWARD] [QUAL]; [AWARD] [QUAL]. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-GROW-A2-W04. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Sort these: weather today; climate over many years. Point to the long-term idea. Staff: Offer a predictable seat or standing spot and the first visual cue. No public test.

Starter (3 min): Think back. Say, point or write one answer for each question. Staff: Ask three low-stakes retrieval questions; prior coverage must be checked.

I Do (5 min): I order the chain: burn fuel → more carbon dioxide → more heat stays → warming. Cutting emissions acts near the start. A flood defence helps with an impact. Staff: Model the full reasoning step, pre-teach four words and ask the show-me question.

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Staff answers continued

We Do (7 min): The burn fuel and impact cards are in place. Sort the missing gas and warming cards. Explain the link. Staff: Show the part-done example, then allow paired talk before an answer. One-minute reset inside this stage.

We Do 2 (5 min): Choose what follows burning more fossil fuel in our warming chain. Everyone answers together. Staff: All pupils answer the four-option hinge; use the decision rule, not a score.

Independent (8 min): Build a cause-to-impact chain. Compare an emissions cut with help for an impact. Give one reason for each. Staff: One adult supports agreed pupils, the other holds the group; swap across the week. One-minute reset inside this stage.

Review (4 min): Hear feedback. Move one chain card and say why. Staff: Give a specific verbal comment (VF) and invite one self-edit (//) or read-back (?).

Exit (3 min): Name a cause, an effect and an action. Which link does your action change? Staff: Hear the answer and name how it changes the next example or support. One-minute reset and handoff.

Planned adaptation — cognition learning: Use four large chain cards with the first and last fixed, then remove a cue.

Planned adaptation — communication interaction: Let partners rehearse an M/H because sentence while pointing to the chain.

Planned adaptation — semh: Give a quiet choice of two card roles and a pause signal before the hinge.

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Staff answers continued

Planned adaptation — sensory physical: Provide large high-contrast cards and a stable board; accept voice capture of the ordering.

Planned adaptation — eal: Teach gas, cause and impact with arrows. Plan in any language; use M/H frames and keep the comparison challenge.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: cognitive/metacognitive strategies; semh: flexible grouping; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Build a cause-to-impact chain. Compare an emissions cut with help for an impact. Give one reason for each. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W05. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W05: Consolidate forces, space and climate; answer scientific questions.

If behind — dependency: Pass on today’s checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: □ Yes □ Partly □ No

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Staff answers continued

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Explain that greenhouse gases slow heat leaving Earth. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A bus depot burns more diesel. Which immediate link belongs in its emissions chain?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Sort these: weather today; climate over many years. Point to the long-term idea. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Build a cause-to-impact chain. Compare an emissions cut with help for an impact. Give one reason for each. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

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EAL: Model/shared rehearsal: use climate, carbon dioxide, greenhouse gas, impact with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own [SYS] target on the school's evidence app; the [SYS] carries their [SYS]-linked and [SYS] targets, so one record serves every plan. Follow the pupil's [SYS] where it differs from these routines.

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Current source checks

Arrival: climate over many years is the long-term idea.

Sort in source order: Burn fossil fuel = cause; More greenhouse gas = middle link; Heat risk in some places = impact; Cut fuel use = action.

Step check: Flood defence helps with an impact; cutting fuel use cuts emissions. Preserve both options and their order.

Hinge: first option, More carbon dioxide is released. Parallel case and full diagnoses are retained in source answers.

Ice task: make a prediction, then record actual Start/Later observations in the table. Do not invent results or promise dark/light ice will differ. Same light, equal starting pieces, one common observation time. No heater; trays and spill care. This observation is not proof or a model of the greenhouse effect.

No compulsory personal consumption or travel disclosure. Treat choices as council or shared-system scenarios.

At Review use specific VF and NS+. Receive the pupil response in their own colour. R requires Voice + Audience + NS+ prompt + pupil response, never automated. Keep prompting separate from pupil evidence.

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